LUKE THOMPSON

SOFTWARE ENGINEER · AI TOOLING & VERIFICATION

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Portfolio: xylardark.github.io/Portfolio-Studio/u/admin/citi · GitHub: github.com/XylarDark

PROFESSIONAL SUMMARY

Developer who builds AI-assisted tooling and the checks that keep it honest. Since February 2026 I've directed AI coding agents on HomeWorld, my solo Unreal Engine 5.8 / C++ project, and built Python tooling that verifies their output against real state. I also build in React/TypeScript (Portfolio Studio, Supabase/Postgres) and TypeScript with Vitest (Shift). Earlier: game UI/UX and C# REST integrations (OutFusion), C++ ECU firmware and JS/SQL web (TruckSail), and client software and QA (Evenica).

CORE SKILLS

Languages	TypeScript/JavaScript, Python, SQL, C#, C++, HTML/CSS
Web & data	React, Vite, REST APIs, JSON schemas, Supabase (Postgres), SQL Server, MySQL
AI tooling	Directing AI coding agents with review gates, JSON tool bridges (MCP), verification of agent output
Testing & CI	Vitest, pytest, headless-Chrome checks, GitHub Actions, Git/GitHub, pull-request review

SELECTED PROJECTS

HomeWorld — AI-agent tooling and verification Feb 2026 – present

Independent (solo) · Unreal Engine 5.8, C++, Python · private repository, available on request

- Built an editor automation bridge: agents and scripts send a JSON `{command, args}` request and get structured JSON back (world info, asset checks, a per-row pass/fail probe); new command modules plug in.
- Built checks that refuse a false "done": a headless test runner that hard-fails on five causes of fake passes (unreadable report, wrong engine, zero tests run, stale report, stale binaries), each with a regression test, and a build wrapper with one machine-wide queue so parallel agents can't collide.
- Run a multi-agent workflow (research, design, implementation, testing roles) with written handoff rules and merge gates; source changes wait for my review.
- A tool-capability matrix gives every claim an evidence level from E0 (only claimed in a doc) to E5 (survived a fresh editor). 42 C++ automation test files and 150+ Python test files.

Portfolio Studio — media-résumé site Sep 2026 – present

Independent · React, TypeScript, Vite, Supabase · xylardark.github.io/Portfolio-Studio

- React/TypeScript portfolio product with Studio editing, a Supabase (Postgres) backend and per-employer pages, built with AI coding agents through pull requests I review; headless-Chrome checks at desktop and phone sizes, and GitHub Actions deploys each merge.

Shift — playable shop & delivery game Aug – Sep 2026

Independent · Phaser 3, TypeScript, Vite · Live: xylardark.github.io/ShiftGame

- Shipped a TypeScript browser/PWA game for desktop and phones with a fixed 60 Hz simulation and cached A* routing; 83 Vitest test files in the public repo.

EXPERIENCE

Game Developer · OutFusion · Hamilton, ON Feb 2022 – Dec 2022

- Implemented game UI/UX from the design docs; built and tested C# REST API integrations for player data; integrated and tested game UI across mobile and desktop; partnered on bug verification and release quality.

Software Developer · TruckSail · Ancaster, ON Feb 2020 – Jan 2021

- Built JS/Bootstrap UIs and optimized SQL; wrote C++ firmware for truck ECU devices (STM32, CAN bus) with defect finding for reliability; maintained GitHub workflows and did cross-browser QA.

Software Developer & QA Tester · Evenica · Burlington, ON 2018 – Sep 2019

- Built and tested client software (Andrew Peller, Young's Market Company, MM Realty International): T-SQL, C#, Linux, and Laravel (MySQL, PHP, JavaScript/jQuery, Vue.js).

Delivery Driver (Uber, retail delivery) · Hamilton, ON 2022 – 2026

EDUCATION

Diploma, Computer Systems Technology — Software Development · Mohawk College · Hamilton, ON · 2019

Valid Ontario Class G licence · Open to hybrid in Mississauga (commuting from Hamilton)